Homework 2

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Exercise 1

$$\begin{bmatrix} 1 & 2 & 7 \\ 3 & 5 & -1 \\ 6 & 1 & 4 \end{bmatrix}$$

Putting the matrix in row echelon form using Gaussian elimination:

$$\begin{array}{ll} \begin{bmatrix} 1 & 2 & 7 \\ 3 & 5 & -1 \\ 6 & 1 & 4 \end{bmatrix} & \begin{array}{ll} -3 & -6 \\ + & \\ & + \end{array} \\ \begin{bmatrix} 1 & 2 & 7 \\ 0 & -1 & -22 \\ 0 & -11 & -38 \end{bmatrix} & \begin{array}{l} \\ -11 \\ + \end{array} \\ \begin{bmatrix} 1 & 2 & 7 \\ 0 & -1 & -22 \\ 0 & 0 & 204 \end{bmatrix} & \end{array}$$

The LU factorization of the matrix is then:

$$\begin{bmatrix} 1 & 2 & 7 \\ 3 & 5 & -1 \\ 6 & 1 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 6 & 11 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 7 \\ 0 & -1 & -22 \\ 0 & 0 & 204 \end{bmatrix}$$

Exercise 2

Let $A \in \mathbb{C}^{n\times n}$ and $b, x \in \mathbb{C}^n$. Solving for x in $Ax = b$ can be done using only real-valued matrices and vectors as follows:

Note that A can be decomposed into two real-valued matrices as $A_{\mathrm{Re}} + iA_{\mathrm{Im}}$. Similarly, b and x can be decomposed as $b_{\mathrm{Re}} + ib_{\mathrm{Im}}$ and $x_{\mathrm{Re}} + ix_{\mathrm{Im}}$. Then $Ax = b$ can be rewritten as:

$$\begin{aligned} (A_{\mathrm{Re}} + iA_{\mathrm{Im}})(x_{\mathrm{Re}} + ix_{\mathrm{Im}}) &= b_{\mathrm{Re}} + ib_{\mathrm{Im}} \\ (A_{\mathrm{Re}}x_{\mathrm{Re}} - A_{\mathrm{Im}}x_{\mathrm{Im}}) + i(A_{\mathrm{Re}}x_{\mathrm{Im}} + A_{\mathrm{Im}}x_{\mathrm{Re}}) &= \end{aligned}$$

This can be modelled as the following linear system, which uses only real-valued matrices:

$$\begin{bmatrix} A_{\mathrm{Re}} & -A_{\mathrm{Im}} \\ A_{\mathrm{Im}} & A_{\mathrm{Re}} \end{bmatrix} \begin{bmatrix} x_{\mathrm{Re}} \\ x_{\mathrm{Im}} \end{bmatrix} = \begin{bmatrix} b_{\mathrm{Re}} \\ b_{\mathrm{Im}} \end{bmatrix}$$

Exercise 3

Let L be an invertible 2×2 lower triangular matrix. It then has the form:

$$L = \begin{bmatrix} l_{1,1} & 0 \\ l_{2,1} & l_{2,2} \end{bmatrix}$$

Its inverse L^{-1} then has the form:

$$L^{-1} = (l_{1,1} l_{2,2})^{-1} \begin{bmatrix} l_{2,2} & 0 \\ -l_{2,1} & l_{1,1} \end{bmatrix}$$

L^{-1} is then also a lower triangular matrix. Now suppose that for all invertible $n \times n$ lower triangular matrices, the inverse is also lower triangular. Then let L be an invertible $(n+1) \times (n+1)$ lower triangular matrix. It then has the form:

$$L = \begin{bmatrix} L' & 0 \\ L'' & l_{n+1,n+1} \end{bmatrix}$$

where L' is an invertible $n \times n$ lower triangular matrix and L'' is a $1 \times n$ matrix. Its inverse L^{-1} then has the form:

$$L^{-1} = \begin{bmatrix} (L')^{-1} & 0 \\ -l_{n+1,n+1}^{-1} L'' L'^{-1} & l_{n+1,n+1}^{-1} \end{bmatrix}$$

L^{-1} is then also a lower triangular matrix. By induction, for all invertible lower triangula matrices, the inverse is lower triangular.

Exercise 4

$$M = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$$

Part (a)

Consider the matrix product:

$$\begin{bmatrix} I & 0 \\ CA^{-1} & I \end{bmatrix} \begin{bmatrix} A & 0 \\ 0 & D - CA^{-1}B \end{bmatrix} \begin{bmatrix} I & A^{-1}B \\ 0 & I \end{bmatrix}$$

It can be simplified as follows:

$$\begin{aligned} \begin{bmatrix} I & 0 \\ CA^{-1} & I \end{bmatrix} \begin{bmatrix} A & 0 \\ 0 & D - CA^{-1}B \end{bmatrix} \begin{bmatrix} I & A^{-1}B \\ 0 & I \end{bmatrix} &= \begin{bmatrix} A & 0 \\ C & D - CA^{-1}B \end{bmatrix} \begin{bmatrix} I & A^{-1}B \\ 0 & I \end{bmatrix} \\ &= \begin{bmatrix} A & B \\ C & D \end{bmatrix} = M \end{aligned}$$

Part (b)

Let A have the LU factorization L_1U_1 and $D-CA^{-1}B$ have the LU factorization L_2U_2. M can then be rewritten as:

$$\begin{aligned} M &= \begin{bmatrix} I & 0 \\ CA^{-1} & I \end{bmatrix} \begin{bmatrix} L_1U_1 & 0 \\ 0 & L_2U_2 \end{bmatrix} \begin{bmatrix} I & A^{-1}B \\ 0 & I \end{bmatrix} \\ &= \begin{bmatrix} I & 0 \\ CA^{-1} & I \end{bmatrix} \begin{bmatrix} L_1 & 0 \\ 0 & L_2 \end{bmatrix} \begin{bmatrix} U_1 & 0 \\ 0 & U_2 \end{bmatrix} \begin{bmatrix} I & A^{-1}B \\ 0 & I \end{bmatrix} \\ &= \begin{bmatrix} L_1 & 0 \\ CA^{-1}L_1 & L_2 \end{bmatrix} \begin{bmatrix} U_1 & U_1A^{-1}B \\ 0 & U_2 \end{bmatrix} \end{aligned}$$

The matrices are lower triangular and upper triangular, respectively. The LU factorization of M is then:

$$M = \begin{bmatrix} L_1 & 0 \\ CA^{-1}L_1 & L_2 \end{bmatrix} \begin{bmatrix} U_1 & U_1A^{-1}B \\ 0 & U_2 \end{bmatrix}$$

Part (c)

The following recursive algorithm finds the LU factorization of a matrix M: LU factorization vis Gaussian elimination has a time complexity of $O(n^3)$. Block LU factorization also has a time complexity of $O(n^3)$, mainly due to having to perform matrix inversion.

Algorithm 1 LU

$n \leftarrow \text{size}(M)$

$\begin{bmatrix} A & B \\ C & D \end{bmatrix} \leftarrow M$

$L_1, U_1 \leftarrow \text{LU}(A)$

$L_2, U_2 \leftarrow \text{LU}(D - CA^{-1}B)$

$L \leftarrow \begin{bmatrix} L_1 & 0 \\ CA^{-1}L_1 & L_2 \end{bmatrix}$

$U \leftarrow \begin{bmatrix} U_1 & U_1A^{-1}B \\ 0 & U_2 \end{bmatrix}$

Exercise 5

Suppose A has two different LU factorizations, L_1U_1 and L_2U_2. Then:

$$\begin{aligned} L_1U_1 &= L_2U_2 \\ L_2^{-1}L_1U_1 &= U_2 \\ L_2^{-1}L_1 &= U_2U_1^{-1} \end{aligned}$$

Since the product of two unit lower triangular matrices is unit lower triangular, and likewise for upper triangular matrices, $L_2^{-1}L_1 = U_2U_1^{-1}$ must be both upper and lower triangular. The only matrix that is both upper and lower triangular is I, which means that $L_2^{-1}L_1 = U_2U_1^{-1} = I$. For this to be true, $L_1 = L_2$ and $U_1 = U_2$; consequently, each matrix permits a unique LU factorization.

Exercise 6

Part (a)

Suppose A is symmetric positive definite. Let A have the LU factorization LU. Since A is symmetric $A = A^T$, and so $A = LU = U^T L^T$. U^T is lower triangular; for some diagonal D, it can be written as ${U'}^T D$, where ${U'}^T$ is unit lower triangular. ${L'}^T = DL^T$ is then upper triangular, so ${U'}^T {L'}^T$ is also a LU factorization of A. By the uniqueness of the LU factorization, $L = {U'}^T$ and $U = {L'}^T$. Since A is positive definite, for any $x \neq 0$:

$$\begin{aligned} x^T Ax &= x^T (LDL^T)x \\ &= (L^T x)^T D(L^T x) \\ &= y^T Dy > 0 \end{aligned}$$

For this to be the case, the diagonal entries of D must be positive. We can then define D' such that ${D'}^2 = D$. Then:

$$\begin{aligned} A &= LDL^T \\ &= L{D'}^2 L^T \\ &= (LD')({D'}^T L^T) \\ &= (LD')(LD')^T \end{aligned}$$

Since L is lower riangular and D' is diagonal, LD' is lower triangular.

Suppose $A = LL^T$ for some lower triangular L. Then:

$$\begin{aligned} A^T &= (LL^T)^T \\ &= LL^T = A \end{aligned}$$

Therefore, A is symmetric. Further, for $x \neq 0$:

$$\begin{aligned} x^T (LL^T)x &= (L^T x)^T (L^T x) \\ &= ||L^T x||^2 > 0 \end{aligned}$$

Therefore A is positive definite, and so it is symmetric positive definite.

Part (b)

```
function L = cholesky(A)
    L = A;
    n = size(A, 1);
    for k = 1:n
        for i = 1:k - 1
            s = 0;
            for j = 1:i - 1
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            s = s + L(i, j) * L(k, j);
        end
        L(k, i) = (L(k, i) - s) / L(i, i);
    end
    v = 0;
    for j = 1:k - 1
        v = v + L(k, j) ^ 2;
    end
    L(k, k) = sqrt(L(k, k) - v);
    for i = k+1:n
        L(k, i) = 0;
    end
  end
end
```

Part (c)

For $n = 3$, A and L are:

$$A = \begin{bmatrix} 1.0000 & 0.5000 & 0.3333 \\ 0.5000 & 0.3333 & 0.2500 \\ 0.3333 & 0.2500 & 0.2000 \end{bmatrix}$$
$$L = \begin{bmatrix} 1.0000 & 0.0000 & 0.0000 \\ 0.5000 & 0.2887 & 0.0000 \\ 0.3333 & 0.2887 & 0.0745 \end{bmatrix}$$

For $n = 4$, A and L are:

$$A = \begin{bmatrix} 1.0000 & 0.5000 & 0.3333 & 0.2500 \\ 0.5000 & 0.3333 & 0.2500 & 0.2000 \\ 0.3333 & 0.2500 & 0.2000 & 0.1667 \\ 0.2500 & 0.2000 & 0.1667 & 0.1429 \end{bmatrix}$$
$$L = \begin{bmatrix} 1.0000 & 0.0000 & 0.0000 & 0.0000 \\ 0.5000 & 0.2887 & 0.0000 & 0.0000 \\ 0.3333 & 0.2887 & 0.0745 & 0.0000 \\ 0.2500 & 0.2598 & 0.1118 & 0.0189 \end{bmatrix}$$

For $n = 5$, A and L are:

$$A = \begin{bmatrix} 1.0000 & 0.5000 & 0.3333 & 0.2500 & 0.2000 \\ 0.5000 & 0.3333 & 0.2500 & 0.2000 & 0.1667 \\ 0.3333 & 0.2500 & 0.2000 & 0.1667 & 0.1429 \\ 0.2500 & 0.2000 & 0.1667 & 0.1429 & 0.1250 \\ 0.2000 & 0.1667 & 0.1429 & 0.1250 & 0.1111 \end{bmatrix}$$
$$L = \begin{bmatrix} 1.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 \\ 0.5000 & 0.2887 & 0.0000 & 0.0000 & 0.0000 \\ 0.3333 & 0.2887 & 0.0745 & 0.0000 & 0.0000 \\ 0.2500 & 0.2598 & 0.1118 & 0.0189 & 0.0000 \\ 0.2000 & 0.2309 & 0.1278 & 0.0378 & 0.0048 \end{bmatrix}$$

Exercise 7

Let $||\cdot||$ be a vector norm on $\mathbb{R}^n$ and let $A \in \mathbb{R}^{n\times n}$ such that $\text{rank}(A) = n$. Let $||x||_A = ||Ax||$. Then, for $x, y \in \mathbb{R}^n$:

Since $||\cdot||$ is a vector norm, $||x||_A = ||Ax|| > 0$ and $||\mathbf{0}|| = ||A\mathbf{0}|| = 0$. Further, for $\alpha \in \mathbb{R}$, $||\alpha x||_A = ||A(\alpha x)|| = |\alpha| \cdot ||Ax|| = |\alpha| \cdot ||x||_A$. Finally, $||A(x+y)|| = ||Ax + Ay|| \leq ||Ax|| + ||Ay||$, so $||x+y||_A \leq ||x||_A + ||y||_A$. Therefore $||x||_A$ is a vector norm on $\mathbb{R}^n$.

Exercise 8

Part (a)

Let A, x, $\hat{x}$, b, $\hat{b}$ such that $Ax = b$, $A\hat{x} = \hat{b}$, and A is invertible. Then:

$$\begin{aligned}
A(x - \hat{x}) &= b - \hat{b} \\
x - \hat{x} &= A^{-1}(b - \hat{b}) \\
||x - \hat{x}||_\infty &= ||A^{-1}(b - \hat{b})||_\infty \\
&\leq ||A^{-1}||_\infty \cdot ||b - \hat{b}||_\infty
\end{aligned}$$

Additionally, note that:

$$\begin{aligned}
||Ax||_\infty &= ||b||_\infty \\
||A||_\infty \cdot ||x||_\infty &\geq \\
\frac{1}{||x||_\infty} &\leq \frac{||A||_\infty}{||b||_\infty}
\end{aligned}$$

Combining the inequalities:

$$\begin{aligned}
\frac{||x - \hat{x}||_\infty}{||x||_\infty} &\leq ||A||_\infty ||A^{-1}||_\infty \cdot \frac{||b - \hat{b}||_\infty}{||b||_\infty} \\
&\leq \kappa(A) \frac{||b - \hat{b}||_\infty}{||b||_\infty}
\end{aligned}$$

Part (b)

$$\begin{aligned}
A &= \begin{bmatrix} 1 & 1.01 \\ 0.99 & 1 \end{bmatrix} \\
b &= \begin{bmatrix} 2.01 \\ 1.99 \end{bmatrix} \\
\hat{b} &= \begin{bmatrix} 2 \\ 2 \end{bmatrix}
\end{aligned}$$

The matrix A is very close to singular; the inverse A^{-1} is:

$$\begin{aligned}
A^{-1} &= \frac{1}{1 \cdot 1 - 1.01 \cdot 0.99} \begin{bmatrix} 1 & -1.01 \\ -0.99 & 1 \end{bmatrix} \\
&= \begin{bmatrix} 10000 & -10100 \\ -9900 & 10000 \end{bmatrix}
\end{aligned}$$

While $||A||_\infty = 2.01$, $||A^{-1}||_\infty = 20100$ and so $\kappa(A) = 40401$. This indicates that the relative error in x can be up to 40401 times the relative error in b. Even for a small perturbation of 0.01 in b, this can result in a very large perturbation in A. Computing x and $\bar{x}$:

$$x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\hat{x} = \begin{bmatrix} -200 \\ 200 \end{bmatrix}$$

While the actual perturbation in x is not as large as the upper bound, it is still hundreds of times larger than the perturbation in b.